

Database Systems

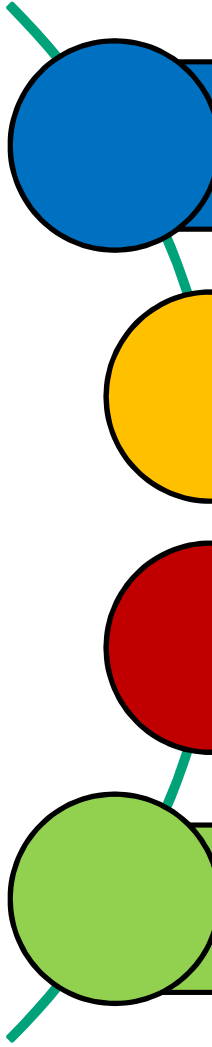
CSE 215

Lecture 05: SQL

2022

Aggregation and Grouping

Outline



Mathematical operators and functions

Aggregations (6.4.3 – 6.4.6)

Having clause

Examples, examples, examples...

Formulas in SQL

So far, only the exact values have been selected from database tables.

It is possible to use formulas where simple literals or column references can appear
(SELECT, WHERE, SET, VALUES used in INSERT, HAVING):

```
SELECT (formula on column name) FROM ...
```

Formula can be created using an *infix* operator:

Function	Definition

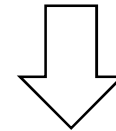
value1 + value2	Addition
value1 - value2	Subtraction
value1 * value2	Multiplication
value1 / value2	Division

Find day-wise sale of each products

Purchase

Product	Day	Price	Quantity
Bagel	10/21/2005	1	20
Banana	10/03/2005	0.5	10
Banana	10/10/2005	1	10
Bagel	10/25/2005	1.50	20

```
SELECT product, day, price*quantity AS Sales
FROM Purchase
```



Product	Day	Sales
Bagel	10/21/2005	20
Banana	10/03/2005	5
Banana	10/10/2005	10
Bagel	10/25/2005	30

NULLS in SQL

- Whenever we don't have a value, we can put a NULL
- Can mean many things:
 - Value does not exist
 - Value exists but is unknown
 - Value not applicable
 - Etc.
- The schema specifies for each attribute whether it can be null (*nullable* attribute) or not
- How does SQL query cope with tables that have NULLs?

Null Values

Any NULL in a formula will cause a NULL result, **not a value of 0**.

Marks

student	mark		student	mark	mark/3
Paul Smith	43		Paul Smith	43	14.33333333
Rachel Sewell	57	→	Rachel Sewell	57	19
Helen Treacy	32		Helen Treacy	72	24
Pinder Surree	NULL		Pinder Surree		

```
SELECT student, mark/3
FROM Marks
```

Null Values

- If $x = \text{NULL}$ then $x = \text{'Joe'}$ is **UNKNOWN**
- In SQL there are three boolean values:

FALSE	=	0
UNKNOWN	=	0.5
TRUE	=	1

A tuple/row having only TRUE
in WHERE clause is passed to
SELECT

3	SELECT	<attributes>
1	FROM	<one or more relations>
2	WHERE	<conditions>

Null Values

- $C1 \text{ AND } C2 = \min(C1, C2)$
- $C1 \text{ OR } C2 = \max(C1, C2)$
- $\text{NOT } C1 = 1 - C1$

```
SELECT *  
FROM Person  
WHERE (age < 25) AND  
      (height > 6 OR weight > 190)
```

Tuple with the
following values
age=20
height=NULL
weight=200

Will it be in the result?

Rule in SQL: include only tuples that yield TRUE

Answer: YES

Null Values

- $C1 \text{ AND } C2 = \min(C1, C2)$
- $C1 \text{ OR } C2 = \max(C1, C2)$
- $\text{NOT } C1 = 1 - C1$

```
SELECT *  
FROM Person  
WHERE (age < 25) AND  
      (height > 6 OR weight > 190)
```

Tuple with the
following values
age=NULL
height=5
weight=200
Will it be in the result?

Answer: NO

Null Values

Unexpected behavior:

```
SELECT *  
FROM Person  
WHERE age < 25 OR age >= 25
```

Some Persons are not included !

Null Values

Can test for NULL explicitly:

- x IS NULL
- x IS NOT NULL

```
SELECT *  
FROM Person  
WHERE age < 25 OR age >= 25 OR age IS NULL
```

Now it includes all Persons

Aggregation

```
SELECT AVG(price)
FROM Product
WHERE maker='Toyota'
```

```
SELECT COUNT(*)
FROM Product
WHERE year > 1995
```

SQL supports several aggregation operations:

sum, count, min, max, avg

All aggregations apply to a single attribute (multiple attributes can be combined in a formula) [Except COUNT(*)]

NULL values are ignored [Except COUNT(*)]

Aggregation: Count

COUNT applies to duplicates, unless otherwise stated:

```
SELECT COUNT(category)
FROM   Product
WHERE  year > 1995
```

same as COUNT(*)

We probably want:

```
SELECT COUNT(DISTINCT category)
FROM   Product
WHERE  year > 1995
```

More Examples

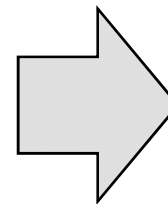
Purchase

Find total sales

Product	Day	Price	Quantity
Bagel	10/21/2005	1	20
Banana	10/03/2005	0.5	10
Banana	10/10/2005	1	10
Bagel	10/25/2005	1.50	20

Purchase

```
SELECT SUM(price * quantity)
FROM Purchase
```



65 (= 20+5+10+30)

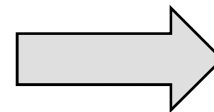
Simple Aggregations

Purchase

Find total sales of product Bagel

Product	Day	Price	Quantity
Bagel	10/21/2005	1	20
Banana	10/03/2005	0.5	10
Banana	10/10/2005	1	10
Bagel	10/25/2005	1.50	20

```
SELECT SUM(price * quantity)
FROM Purchase
WHERE product = 'bagel'
```



50 (= 20+30)

Grouping and Aggregation

Find total sales after 10/04/2005 **per** product.

Product	Day	Price	Quantity
Bagel	10/21/2005	1	20
Banana	10/03/2005	0.5	10
Banana	10/10/2005	1	10
Bagel	10/25/2005	1.50	20

Purchase

```
SELECT    product, SUM(price*quantity) AS "Total Sales"  
FROM      Purchase  
WHERE     day > '04-OCT-05'  
GROUP BY product
```

Let's see what this means...

Grouping and Aggregation

Find total sales after 10/04/2005 **per** product.

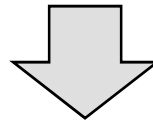
```
SELECT    product, SUM(price*quantity) AS “Total Sales”  
FROM      Purchase  
WHERE     day > '04-OCT-05'  
GROUP BY  product
```

1. Compute the **FROM** and **WHERE** clauses.
2. Group by the attribute(s) in the **GROUP BY**
3. Compute the **SELECT** clause on grouped attributes and aggregates.

1. FROM-WHERE

Find total sales after 10/04/2005 **per** product.

Product	Day	Price	Quantity
Bagel	10/21/2005	1	20
Banana	10/03/2005	0.5	10
Banana	10/10/2005	1	10
Bagel	10/25/2005	1.50	20



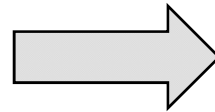
Product	Day	Price	Quantity
Bagel	10/21/2005	1	20
Banana	10/10/2005	1	10
Bagel	10/25/2005	1.50	20

2. GROUP BY

Product	Day	Price	Quantity
Bagel	10/21/2005	1	20
Bagel	10/25/2005	1.50	20
Banana	10/10/2005	1	10

3. SELECT

Product	Day	Price	Quantity
Bagel	10/21/2005	1	20
Bagel	10/25/2005	1.50	20
Banana	10/10/2005	1	10



Product	TotalSales
Bagel	50
Banana	10

```
SELECT    product, SUM(price*quantity) AS "Total Sales"  
FROM      Purchase  
WHERE     day > '04-OCT-05'  
GROUP BY product
```

HAVING Clause

Find all products and their total sales which were sold at least 30 pieces after 10/04/2005.

```
SELECT    product, SUM(price * quantity)
FROM      Purchase
WHERE     day > '04-OCT-05'
GROUP BY  product
HAVING    SUM(quantity) >= 30
```

HAVING clause contains conditions on aggregates.

General form of Grouping and Aggregation

SELECT S
FROM $R_1, \dots, R_n$
WHERE C1
GROUP BY $a_1, \dots, a_k$
HAVING C2



S = may contain attributes $a_1, \dots, a_k$ and/or any aggregates but NO OTHER ATTRIBUTES

C1 = is any condition on the attributes in $R_1, \dots, R_n$

C2 = is any condition on aggregate expressions

General form of Grouping and Aggregation

```
SELECT  S  
FROM    R1,...,Rn  
WHERE   C1  
GROUP BY a1,...,ak  
HAVING  C2
```

Evaluation steps:

1. Evaluate FROM-WHERE, apply condition C1
2. Group by the attributes $a_1, \dots, a_k$
3. Apply condition C2 to each group (may have aggregates)
4. Compute aggregates in S and return the result

Product (maker, model, type)
PC (model, speed, ram, hd, price)
Laptop (model, speed, ram, hd, screen, price)

Solutions
on next
slides

Find the manufacturers that make at least three different models of PCs

Find all manufacturers that produce all three types of products (pc, printer, laptop)

Find all manufacturers that produce only one product type and more than one model

Find the average speed of PCs

Find the average price of PCs made by manufacturer 'A'

Find for each speed of PC above 2.0, the average price

Find for each manufacturer who sell's PCs the maximum price of a PC

Product (maker, model, type)

PC (model, speed, ram, hd, price)

Laptop (model, speed, ram, hd, screen, price)

Find the manufacturers that make at least three different models of PCs

Product (maker, model, type)
PC (model, speed, ram, hd, price)
Laptop (model, speed, ram, hd, screen, price)

Find the manufacturers that make at least three different models of PCs

```
SELECT    maker
FROM      Product
WHERE     type = 'pc'
GROUP BY  maker
HAVING    COUNT(model) >= 3
```

Product (maker, model, type)
PC (model, speed, ram, hd, price)
Laptop (model, speed, ram, hd, screen, price)

Find all manufacturers that produce all three types of products (pc, printer, laptop)

```
SELECT    maker
FROM      Product
GROUP BY  maker
HAVING COUNT(DISTINCT type) = 3
```

Product (maker, model, type)
PC (model, speed, ram, hd, price)
Laptop (model, speed, ram, hd, screen, price)

Find all manufacturers that produce only one product type
and more than one model

```
SELECT    maker
FROM      Product
GROUP BY  maker
HAVING    COUNT(DISTINCT type) = 1
AND       COUNT(model) > 1
```

Product (maker, model, type)
PC (model, speed, ram, hd, price)
Laptop (model, speed, ram, hd, screen, price)

Find the average speed of PCs

```
SELECT    AVG(speed)
FROM      PC
```

Product (maker, model, type)
PC (model, speed, ram, hd, price)
Laptop (model, speed, ram, hd, screen, price)

Find the average price of PCs made by manufacturer 'A'

SELECT	AVG(price)
FROM	Product, PC
WHERE	Product.model = PC.model
AND	maker = 'A'

Product (maker, model, type)
PC (model, speed, ram, hd, price)
Laptop (model, speed, ram, hd, screen, price)

Find for each speed of PC above 2.0, the average price

```
SELECT    speed, AVG(price)
FROM      PC
WHERE     speed > 2.0
GROUP BY  speed
```


Product (maker, model, type)
PC (model, speed, ram, hd, price)
Laptop (model, speed, ram, hd, screen, price)

Find for each manufacturer who sell's PCs the
maximum price of a PC

```
SELECT    maker, MAX(price)
FROM      PC, Product
WHERE     PC.model = Product.model
GROUP BY  maker
```